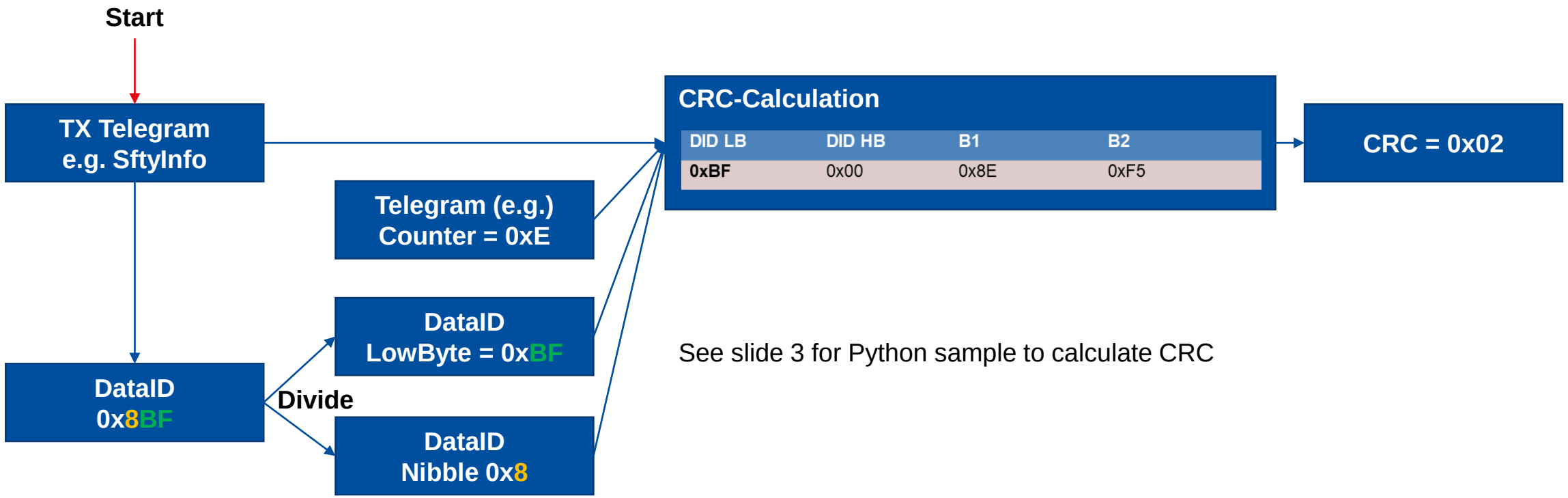
Confidential

E2E Communication

Example SftyInfo telegram

Webasto

Calculation sequence: Example for SftyInfo message



See slide 3 for Python sample to calculate CRC

Take DataID dependent on message (e.g. SftyInfo):
VIC_Gen1_0_Veh_CAN_E2E_DataID.xlsx

Python sample script

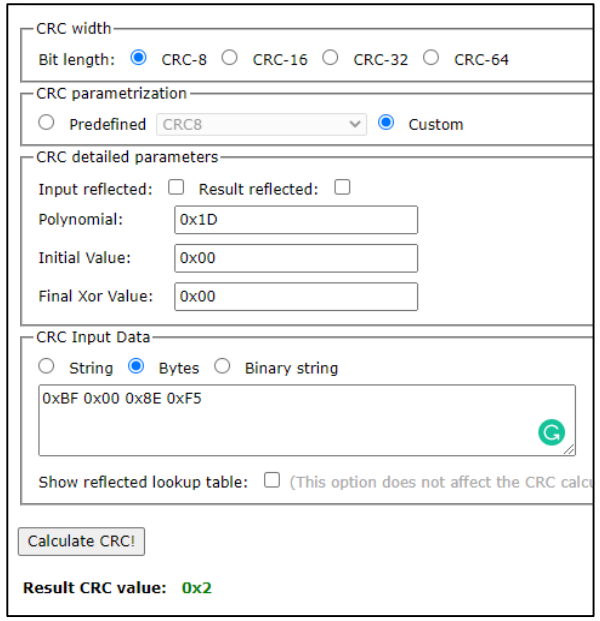
Python script to calculate E2E CRC P1C of SftyInfo with DataID 0x8BF and counter value:

```
table =
[0x00,0x1D,0x3A,0x27,0x74,0x69,0x4E,0x53,0xE8,0xF5,0xD2,0xCF,0x9C,0x81,0xA6,0xBB,0xCD,0xD0,0xF7,0xEA,0xB9,0xA4,0x83,0x9E,0x25,0x38,0x1F,0x02,0x51,0x4C,0x6B,0x76,0x87,0x9A,0xBD,0xA0
,0xF3,0xEE,0xC9,0xD4,0x6F,0x72,0x55,0x48,0x1B,0x06,0x21,0x3C,0x4A,0x57,0x70,0x6D,0x3E,0x23,0x04,0x19,0xA2,0xBF,0x98,0x85,0xD6,0xCB,0xEC,0xF1,0x13,0x0E,0x29,0x34,0x67,0x7A,0x5D,0x40
,0xFB,0xE6,0xC1,0xDC,0x8F,0x92,0xB5,0xA8,0xDE,0xC3,0xE4,0xF9,0xAA,0xB7,0x90,0x8D,0x36,0x2B,0x0C,0x11,0x42,0x5F,0x78,0x65,0x94,0x89,0xAE,0xB3,0xE0,0xFD,0xDA,0xC7,0x7C,0x61,0x46,0x5B
,0x08,0x15,0x32,0x2F,0x59,0x44,0x63,0x7E,0x2D,0x30,0x17,0x0A,0xB1,0xAC,0x8B,0x96,0xC5,0xD8,0xFF,0xE2,0x26,0x3B,0x1C,0x01,0x52,0x4F,0x68,0x75,0xCE,0xD3,0xF4,0xE9,0xBA,0xA7,0x80,0x9D
,0xEB,0xF6,0xD1,0xCC,0x9F,0x82,0xA5,0xB8,0x03,0x1E,0x39,0x24,0x77,0x6A,0x4D,0x50,0xA1,0xBC,0x9B,0x86,0xD5,0xC8,0xEF,0xF2,0x49,0x54,0x73,0x6E,0x3D,0x20,0x07,0x1A,0x6C,0x71,0x56,0x4B
,0x18,0x05,0x22,0x3F,0x84,0x99,0xBE,0xA3,0xF0,0xED,0xCA,0xD7,0x35,0x28,0x0F,0x12,0x41,0x5C,0x7B,0x66,0xDD,0xC0,0xE7,0xFA,0xA9,0xB4,0x93,0x8E,0xF8,0xE5,0xC2,0xDF,0x8C,0x91,0xB6,0xAB
,0x10,0x0D,0x2A,0x37,0x64,0x79,0x5E,0x43,0xB2,0xAF,0x88,0x95,0xC6,0xDB,0xFC,0xE1,0x5A,0x47,0x60,0x7D,0x2E,0x33,0x14,0x09,0x7F,0x62,0x45,0x58,0x0B,0x16,0x31,0x2C,0x97,0x8A,0xAD,0xB0
,0xE3,0xFE,0xD9,0xC4]
```

```
crc = 0x00

data = [0xBF, 0x00, 0x8E, 0xF5]

for i in range(len(data)):

                crc = table[(crc^data[i])]

print(hex(crc))
```

See CRC calculator online script:

http://www.sunshine2k.de/coding/javascript/crc/crc_js.html

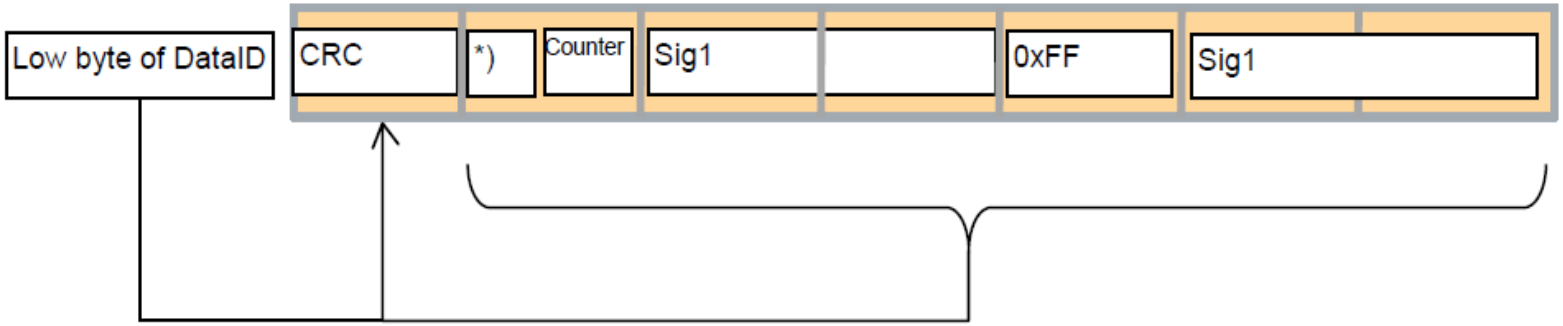


CRC calculation

Cyclic redundancy check (CRC): An 8-bit CRC is explicitly sent with polynomial 0x1D with an initial value 0x00 and a final XOR-value 0x00.

E2E Profile 1 shall use CRC-8-SAE J1850 for CRC calculation. It shall use 0x00 as the start value and XOR value.

The CRC is calculated over the low byte of the DataID and the data itself starting from the byte B1.



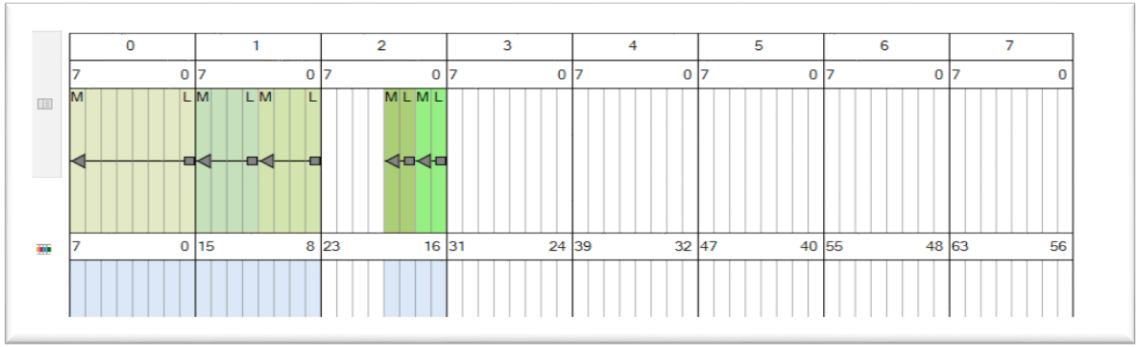
Example of CRC calculation: SftyInfo (VIC) (1/2)

Input documents :

- VIC_Gen1_0_VEH_CAN.arxml
- VIC_Gen1_0_Veh_CAN_E2E_DataID.xlsx

SftyInfo CAN message layout: The DataID of SftyInfo is **0x8BF** (LowByte: 0xBF, HighByte: 0x08)

Signal	Start Position	Length [Bit]
SftyInfo_Cks	0	8
SftyInfo_Alvctr	8	4
SftyInfo_DIDNibble	12	4
SftyOk	16	2
CrashDetnNeg	18	2



Example of CRC calculation: SftyInfo (VIC) (2/2)

1. The HighByte of DID shall be 0 and the unused bits shall be 1.
2. The signals SftyOK and CrashDetnNeg are set to „1“.

DID LB	DID HB	B1	B2
0xBF	0x00	0x8E	0xF5

3. To calculate the CRC you have to use the function Crc_CalculateCRC8() (see python example) with an initial value 0x00 and a final XOR-value 0x00.
4. The lookup table of Crc-8 with polynom 0x1D is used to facilate the calculation.
5. The calculation of the data „0xBF 0x00 0x8E 0xF5“ using CRC-8-SAE J1850 gives **0x02**

References (1/2)

https://www.autosar.org/fileadmin/user_upload/standards/classic/4-2/AUTOSAR_SWS_E2ELibrary.pdf

https://www.autosar.org/fileadmin/user_upload/standards/classic/4-2/AUTOSAR_SWS_CRCLibrary.pdf

CRC	CRC-8-SAE J1850 - 0x1D (x8 + x4 + x3 + x2 + 1), but with different start and XOR values (both start value and XOR value are 0x00). This CRC is provided by CRC library. Starting with AUTOSAR R4.0, the SAE8 CRC function of the CRC library uses 0xFF as start value and XOR value. To compensate a different behavior of the CRC library, the E2E Library applies additional XOR 0xFF operations starting with R4.0, to come up with 0x00 as start value and XOR value. Note: This CRC polynomial is different from the CRC-polynomials used by FlexRay, CAN and LIN.

References (2/2)

Crc-8 SAE-J1850 0x1D lookup table.

```
SCrc_Table8[256] =
{   0x00U, 0x1DU, 0x3AU, 0x27U, 0x74U, 0x69U, 0x4EU, 0x53U, 0xE8U, 0xF5U, 0xD2U,
    0xCFU, 0x9CU, 0x81U, 0xA6U, 0xBBU,
    0xCDU, 0xD0U, 0xF7U, 0xEAU, 0xB9U, 0xA4U, 0x83U, 0x9EU, 0x25U, 0x38U, 0x1FU,
    0x02U, 0x51U, 0x4CU, 0x6BU, 0x76U, 0x87U, 0x9AU, 0xBDU, 0xA0U, 0xF3U, 0xEEU,
    0xC9U, 0xD4U, 0x6FU, 0x72U, 0x55U, 0x48U, 0x1BU, 0x06U, 0x21U, 0x3CU, 0x4AU,
    0x57U, 0x70U, 0x6DU, 0x3EU, 0x23U, 0x04U, 0x19U, 0xA2U, 0xBFU, 0x98U, 0x85U,
    0xD6U, 0xCBU, 0xECU, 0xF1U, 0x13U, 0x0EU, 0x29U, 0x34U, 0x67U, 0x7AU, 0x5DU,
    0x40U, 0xFBU, 0xE6U, 0xC1U, 0xDCU, 0x8FU, 0x92U, 0xB5U, 0xA8U, 0xDEU, 0xC3U,
    0xE4U, 0xF9U, 0xAAU, 0xB7U, 0x90U, 0x8DU, 0x36U, 0x2BU, 0x0CU, 0x11U, 0x42U,
    0x5FU, 0x78U, 0x65U, 0x94U, 0x89U, 0xAEU, 0xB3U, 0xE0U, 0xFDU, 0xDAU, 0xC7U,
    0x7CU, 0x61U, 0x46U, 0x5BU, 0x08U, 0x15U, 0x32U, 0x2FU, 0x59U, 0x44U, 0x63U,
    0x7EU, 0x2DU, 0x30U, 0x17U, 0x0AU, 0xB1U, 0xACU, 0x8BU, 0x96U, 0xC5U, 0xD8U,
    0xFFU, 0xE2U, 0x26U, 0x3BU, 0x1CU, 0x01U, 0x52U, 0x4FU, 0x68U, 0x75U, 0xCEU,
    0xD3U, 0xF4U, 0xE9U, 0xBAU, 0xA7U, 0x80U, 0x9DU, 0xEBU, 0xF6U, 0xD1U, 0xCCU,
    0x9FU, 0x82U, 0xA5U, 0xB8U, 0x03U, 0x1EU, 0x39U, 0x24U, 0x77U, 0x6AU, 0x4DU,
    0x50U, 0xA1U, 0xBCU, 0x9BU, 0x86U, 0xD5U, 0xC8U, 0xEFU, 0xF2U, 0x49U, 0x54U,
    0x73U, 0x6EU, 0x3DU, 0x20U, 0x07U, 0x1AU, 0x6CU, 0x71U, 0x56U, 0x4BU, 0x18U,
    0x05U, 0x22U, 0x3FU, 0x84U, 0x99U, 0xBEU, 0xA3U, 0xF0U, 0xEDU, 0xCAU, 0xD7U,
    0x35U, 0x28U, 0x0FU, 0x12U, 0x41U, 0x5CU, 0x7BU, 0x66U, 0xDDU, 0xC0U, 0xE7U,
    0xFAU, 0xA9U, 0xB4U, 0x93U, 0x8EU, 0xF8U, 0xE5U, 0xC2U, 0xDFU, 0x8CU, 0x91U,
    0xB6U, 0xABU, 0x10U, 0x0DU, 0x2AU, 0x37U, 0x64U, 0x79U, 0x5EU, 0x43U, 0xB2U,
    0xAFU, 0x88U, 0x95U, 0xC6U, 0xDBU, 0xFCU, 0xE1U, 0x5AU, 0x47U, 0x60U, 0x7DU,
    0x2EU, 0x33U, 0x14U, 0x09U, 0x7FU, 0x62U, 0x45U, 0x58U, 0x0BU, 0x16U, 0x31U,
    0x2CU, 0x97U, 0x8AU, 0xADU, 0xB0U, 0xE3U, 0xFEU, 0xD9U, 0xC4U};
```